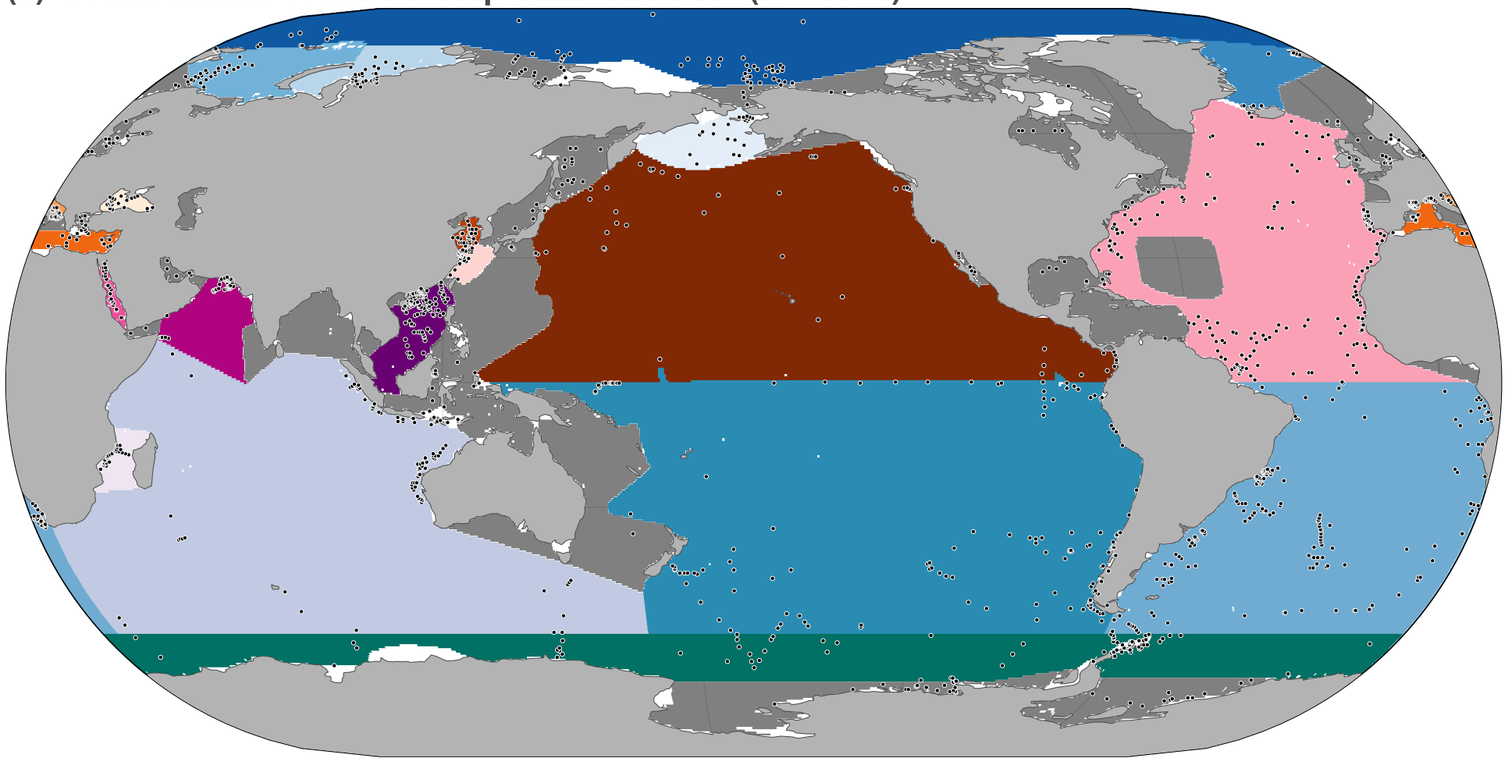
(a) Ocean basins of the coretop calibration set (n = 1518)



Arctic (n = 58)
Barents Sea (n = 28)
Bering Sea (n = 24)
Golfe du Lion (n = 23)
Mediterranean (n = 31)
N. Pacific (n = 46)
N. Atlantic (n = 173)
Arabian Sea (n = 41)
Mozambique Ch. (n = 30)
S. Atlantic (n = 207)
Southern (n = 77)
Greenland Sea (n = 20)
Kara Sea (n = 38)
Black Sea (n = 25)
Adriatic Sea (n = 24)
Yellow Sea (n = 31)
E. China Sea (n = 34)
Red Sea (n = 23)
S. China Sea (n = 117)
Indian (n = 95)
S. Pacific (n = 141)
basins with n < 20 (n = 232)

(b) SST prediction RMSE by ocean basin (n ≥ 20)

21 basins hold 1286 of 1518 sites (85%); black box = lowest RMSE in that basin

| | Arctic | Greenland Sea | Barents Sea | Kara Sea | Bering Sea | Black Sea | Golfe du Lion | Adriatic Sea | Mediterranean | Yellow Sea | N. Pacific | E. China Sea | N. Atlantic | Red Sea | Arabian Sea | S. China Sea | Mozambique Ch. | Indian | S. Atlantic | S. Pacific | Southern | All RMSE | Std of region-RMSE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Linear (Schouten+ 2002) | 9.5 | 6.6 | 4.3 | 8.6 | 5.6 | 5.2 | 5.9 | 5.8 | 5.2 | 5.8 | 3.7 | 3.1 | 4.4 | 9.5 | 4.3 | 5.1 | 2.1 | 3.5 | 3.9 | 2.2 | 7.1 | 5.3 | 2.1 |
| Exponential (Kim+ 2010) | 12.5 | 9.8 | 6.9 | 11.7 | 8.7 | 2.2 | 3.2 | 3.5 | 5.6 | 3.2 | 4.9 | 1.7 | 3.8 | 5.7 | 2.9 | 3.3 | 1.3 | 3.7 | 3.5 | 3.5 | 9.9 | 5.9 | 3.3 |
| BAYSPAR (Tierney+ 2014) | 8.7 | 7.7 | 4.9 | 8.9 | 4.2 | 4.5 | 5.1 | 4.9 | 5.3 | 2.8 | 2.6 | 1.6 | 3.7 | 1.5 | 3.6 | 3.2 | 1.7 | 3.2 | 2.8 | 2.2 | 3.1 | 4.2 | 2.1 |
| TEXAS (T only) | 3.7 | 4.2 | 3.2 | 2.5 | 3.3 | 3.2 | 3.9 | 2.5 | 2.7 | 3.3 | 3.8 | 5.4 | 4.6 | 6.7 | 5.2 | 2.7 | 2.0 | 3.5 | 4.9 | 3.8 | 3.0 | 3.9 | 1.1 |
| TEXAS (T + G23) | 4.6 | 5.2 | 5.1 | 3.6 | 5.1 | 3.7 | 5.3 | 3.3 | 4.4 | 4.0 | 3.3 | 4.4 | 4.3 | 5.2 | 3.3 | 2.5 | 0.9 | 3.2 | 5.2 | 4.5 | 4.2 | 4.2 | 1.1 |
| TEXAS (T + G23 + $NO_3^-$) | 5.0 | 5.7 | 5.3 | 4.1 | 5.4 | 4.3 | 5.0 | 3.3 | 3.0 | 3.9 | 3.6 | 5.4 | 4.2 | 2.1 | 2.5 | 2.8 | 2.2 | 3.5 | 5.4 | 4.8 | 4.7 | 4.3 | 1.1 |

RMSE (°C): 13.5, 12, 10.5, 9, 7.5, 6, 4.5, 3, 1.5, 0